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Comparative Evaluation of YOLOv11n Neck-Level Modifications for Precast Component and PPE Object Detection in Construction Environments

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Abstract

Construction monitoring systems can benefit from automated detection of both structural components and personal protective equipment (PPE). While previous studies focus on single-task applications using YOLOv5-YOLOv10 architectures, this study presents a systematic comparative evaluation of four neck-level architectural modifications within the YOLOv11n framework: Depthwise separable convolutions (DSC) for computational efficiency, multi-scale dilated convolutions (MSDC) for expanded receptive fields, feature refinement interfaces (FRI) for learnable feature adaptation, and dual attention mechanisms (DAM) for enhanced feature discriminability. Controlled experiments were conducted on precast components (3771 images, 6 classes) and PPE (5201 images, 3 classes) datasets. DAM-YOLO achieved the highest performance with mAP@50 of 0.972 (precast) and 0.968 (PPE), while the performance range across all variants spanned 0.942–0.972 (precast) and 0.936–0.968 (PPE). All variants demonstrated robust detection capabilities with mAP@50 ≥ 0.936 and mAP@50–95 spanning 0.760–0.825. The narrow performance differences across variants indicate that different neck-level modifications yield comparable detection accuracy, providing empirical evidence to support architecture selection within these two evaluated construction object detection scenarios.

Keywords: construction; automation; detection; safety; precast; monitoring; deep learning

1. Introduction

Off-site construction, particularly precast concrete manufacturing, has gained prominence due to benefits such as enhanced quality control, accelerated timelines, and improved safety. This paradigm involves fabricating components like beams, columns, and slabs off-site before transporting them for assembly. However, this workflow introduces distinct safety and monitoring challenges across handling, logistics, and on-site worker protection. Traditional monitoring approaches are often labor-intensive and prone to error [1–4]. Furthermore, the accurate, real-time detection of multiple personal protective equipment (PPE) items in complex construction imagery remains a significant challenge [5]. Automated monitoring systems capable of concurrently detecting precast components and PPE therefore offer potential benefits for addressing both productivity and safety requirements.

Recent advances in deep learning-based computer vision have demonstrated progress in automated construction object detection across several task categories, with

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empirical studies reporting competitive performance of YOLO architectures in specific engineering applications. For structural component detection, notable progress has focused on automated rebar counting from UAV imagery, with large-scale models using Faster R-CNN and YOLOv5/v6 achieving 94.61% AP50 at 32 FPS, while the first publicly available labeled UAV dataset was contributed for this purpose [6,7]. Systematic evaluation of traditional augmentation methods using Faster R-CNN and YOLOv10-based Transformer architectures achieved performance improvements of +5.92 pp AP50 but remained focused on this singular structural element [8]. YOLO architectures have additionally demonstrated competitive performance in other construction vision tasks, including window detection, which achieved 89.93–91.53% precision and 90.82–92.73% IoU across multiple architectural configurations [9], providing contextual evidence of broader applicability.

In parallel, PPE-related object detection and worker monitoring research has leveraged newer architectures for distinct sub-tasks. Researchers have developed specialized systems across multiple equipment categories, addressing comprehensive recognition including hardhats, masks, gloves, and safety vests alongside their non-compliance counterparts [10–13]. Recent advances include non-PPE detection using YOLOv8 with Transformer backbones (73.12% AP50), YOLOv10-based approaches achieving 87.32% AP50 for multi-class detection, and safety helmet monitoring demonstrating 94.24% mAP [5,14,15]. These technical foundations enable the transition from manual inspection protocols to continuous automated surveillance systems through existing non-intrusive camera infrastructure without requiring wearable sensors [16,17], while attention mechanisms and lightweight architectures have enabled real-time monitoring capabilities [10,16,18–20]. For construction equipment detection, heavy equipment identification using YOLOv10-Transformer architectures achieved 89.72% AP50 at 35 FPS [21], representing a complementary but distinct task category. Table A1 presents representative studies addressing PPE detection and precast component identification across all categories, revealing several key limitations.

However, several research gaps remain in the literature. The aforementioned studies, while demonstrating the efficacy of modern architectures and comprehensive methodologies, exhibit clear limitations: focus on singular object categories (either rebars, specific PPE items, or equipment), employment of architectures preceding YOLOv11, and lack of integration between component detection and holistic safety monitoring. A controlled evaluation framework that can systematically assess both precast component detection and PPE-related object detection within a consistent experimental protocol is therefore warranted. While general construction object detection has received substantial attention, specialized frameworks for precast concrete components remain limited. Jang et al. [22] developed a precast concrete dataset for automated component recognition yet focused on component identification without concurrent worker safety monitoring. Additionally, annotation heterogeneity across PPE categories—varying terminologies for helmets, vests, and protective equipment—complicates the development of unified detection frameworks [23,24]. Most approaches target individual PPE categories without addressing comprehensive safety compliance across multiple equipment types [12,13,20].

Collectively, the current literature predominantly utilizes YOLOv5, YOLOv8, or YOLOv10 architectures for single-task applications [10,12,22] and typically validates single model configurations for specific tasks rather than conducting controlled comparison of multiple architectural variants within consistent evaluation protocols. This study addresses these methodological gaps through a unified evaluation framework, applying a systematic controlled comparison of four neck-level modifications within the YOLOv11n framework—depthwise separable convolutions (DSC), multi-scale dilated convolutions (MSDC), feature refinement interfaces (FRI), and dual attention mechanisms (DAM). Each

modification adapts established techniques from the literature to the neck-level feature fusion stage. The primary contribution lies not in proposing fundamentally new architectures but in providing empirical evidence to support architecture selection across two construction object detection scenarios: precast component detection and PPE-related object detection, under identical experimental conditions.

This study pursues three objectives: (1) establish controlled evaluation protocols using publicly available datasets for precast concrete component detection and PPE-related object detection; (2) evaluate four neck-level modifications to YOLOv11n under identical training configurations across both construction object detection scenarios; (3) provide empirical evidence to support architecture selection decisions within the two evaluated construction object detection scenarios.

2. Methodology

2.1. Baseline Architecture

YOLOv11n was selected as the foundational model for this study based on its compact architecture. While prior research in this domain has predominantly employed YOLOv5, YOLOv8, or YOLOv10 (see Table A1), YOLOv11 incorporates architectural updates that enhance feature extraction capabilities and processing efficiency. Comparative evaluations report competitive detection performance alongside reduced inference times relative to predecessor architectures [25–27]. The nano (n) variant contains approximately 2.624 million parameters and 6.3 GFLOPs [20], making it among the most compact variants in the YOLOv11 family [28], providing a computationally lean baseline for neck-level architectural comparison.

YOLOv11 represents the latest advancement in the You Only Look Once series, designed for multiple computer vision tasks including object detection, instance segmentation, and image classification [29]. Key architectural innovations include the C3k2 (Convolutional Cross-Connections with Kernel size 2) module for efficient gradient flow and the C2PSA (Cross-Stage Partial Self-Attention) module for enhanced spatial attention, collectively improving feature extraction and contextual understanding [30,31]. The model family spans variants from nano to extra-large, enabling optimization between computational efficiency and detection accuracy [26]

The YOLOv11 architecture, detailed in Algorithm A1, comprises the following three core components.

- Backbone: Extracts hierarchical features using convolutional layers, C3k2 modules with bypass connections, and spatial pyramid pooling fast (SPPF). A key component is the C2PSA module, which utilizes positional self-attention to enhance critical features [29].
- Neck: Fuses multi-scale features through upsampling, concatenation, and further C3k2 processing.
- Head: Performs detection across scales, employing depthwise separable convolutions for efficiency [32] to output bounding boxes, classes, and confidence scores.

This YOLOv11n foundation enables evaluation of targeted architectural modifications while maintaining computational efficiency essential for practical construction monitoring deployment [1,33].

2.2. Evaluated Variants

2.2.1. DSC-YOLO

DSC-YOLO targets parameter reduction in the neck architecture through depthwise separable convolution. The conventional C3k2 blocks at neck layers 13, 16, 19, and 22 are replaced with DWTinyBlock modules. Each DWTinyBlock decomposes standard 3 × 3

convolutions into two sequential operations: a depthwise spatial convolution (3 × 3, groups = C) followed by a pointwise channel mixing operation (1 × 1) [32]. This factorization reduces computational cost from $O(C^2 \cdot k^2)$ to $O(C \cdot k^2 + C^2)$ while preserving feature extraction capability [32,34], as shown in Algorithm A2.

2.2.2. MSDC-YOLO

MSDC-YOLO expands receptive field coverage through multi-scale dilated convolutions [35]. The DilatedContextBlock, replacing C3k2 at neck layers 13, 16, 19, and 22, processes input features through three parallel branches with dilation rates $d \in \{1, 2, 3\}$. Each branch employs 3 × 3 convolutions with different dilation rates, capturing contextual information at local (d = 1), medium-range (d = 2), and global (d = 3) scales. Multi-scale dilated convolutions enable expanded receptive field coverage for contextual feature extraction [35], with applications demonstrated in construction monitoring contexts [36]. The effective receptive fields are $(2d + 1) \times (2d + 1) = \{3 \times 3, 5 \times 5, 7 \times 7\}$ without additional spatial downsampling. The parallel features are concatenated along the channel dimension and fused through a 1 × 1 convolution to produce the final output, as shown in Algorithm A3.

2.2.3. FRI-YOLO

FRI-YOLO preserves the standard YOLOv11n backbone and neck architecture while introducing a lightweight feature refinement interface between the neck outputs and detection head. Three LinearLiteBlock modules are inserted to process the P3, P4, and P5 feature pyramids independently. Each LinearLiteBlock applies a channel-wise transformation through a 1 × 1 convolution (k = 1, s = 1, p = 0) followed by batch normalization and SiLU activation, maintaining channel dimensionality (C_in = C_out). This pre-head interface functions as a trainable layer that adapts feature representations specifically for the detection task without modifying the feature extraction pathway, as shown in Algorithm A4.

2.2.4. DAM-YOLO

DAM-YOLO integrates dual attention mechanisms at neck layers 13, 16, 19, and 22 through TAFHybridBlock modules. Each TAFHybridBlock first applies a 3 × 3 convolutional stem transformation, then processes feature through two parallel attention branches: efficient channel attention (ECA) [37] and squeeze-and-excitation (SE) [38]. The ECA module employs efficient 1D convolution (k = 3) for local cross-channel interaction modeling. Attention mechanisms allow the model to distinguish features more effectively in object detection [37,38], with recent applications in construction object detection scenarios [39]. The SE module uses two fully connected layers with a reduction ratio r = 8 ($C \rightarrow C/8 \rightarrow C$) for global channel-wise feature recalibration. The attended features from both branches are concatenated ($2C$ channels) and fused through a 1 × 1 convolution to the original channel dimension, providing complementary attention that captures both local channel dependencies and global channel statistics, as shown in Algorithm A5.

2.2.5. Comparative Analysis

Table A2 summarizes the architectural specifications of the four evaluated variants. DSC-YOLO achieves a marginal parameter reduction through depthwise separable factorization. MSDC-YOLO expands receptive field coverage through parallel dilated convolutions. FRI-YOLO introduces minimal-overhead feature refinement through 1 × 1 convolutions. DAM-YOLO integrates dual attention mechanisms through TAFHybridBlock modules. All modifications target the neck's multi-scale, feature fusion stage (layers 13, 16, 19, 22) while preserving the standard YOLOv11n backbone and detection head.

The selection of the four specific neck-level modifications—DSC, MSDC, FRI, DAM—is driven by addressing distinct challenges in automated construction monitoring, building upon well-established architectural principles:

- DSC-YOLO targets parameter reduction for deployment in resource-constrained environments, adapting the depthwise separable convolution paradigm [32,34] to reduce parameters and FLOPs in the feature fusion neck, a critical consideration for real-time processing on resource-constrained site hardware.
- MSDC-YOLO addresses the need for multi-scale contextual awareness, crucial for detecting objects of vastly different sizes (e.g., large precast beams and small safety helmets). It incorporates parallel dilated convolutions [35] to expand the effective receptive field without additional pooling, helping to capture context around occluded or partially visible objects common on construction sites.
- FRI-YOLO is designed as a minimal-intervention baseline to evaluate the impact of a simple, learnable feature adaptation layer before the detection head. It tests the hypothesis of whether a lightweight, post-neck refinement can compensate for feature misalignments without altering the core fusion pathway.
- DAM-YOLO integrates two complementary attention mechanisms—ECA [37] and SE [38]—to enhance feature discriminability. This design aims to improve the model's focus on informative spatial and channel features amidst cluttered construction backgrounds, a recurring challenge noted in prior studies [39].

These modifications integrate established techniques [32,35,37,38] into the YOLOv11n neck architecture. Methodologically, this study focuses on a comparative evaluation of complete architectural variants to assess their integrated performance in applied scenarios, rather than conducting an ablation analysis of individual modules. The primary contribution of this comparative study is the systematic evaluation of these distinct enhancement strategies within a unified, modern framework applied to dual-domain construction monitoring, an area where such controlled architectural comparisons are limited.

2.3. Dataset Overview

This study employs two publicly available datasets to evaluate the evaluated variants across distinct construction monitoring scenarios. Table 1 provides an overview of the datasets, which target precast component detection and worker and PPE detection, respectively.

Table 1. Overview of datasets for construction scene.

Task	Dataset	Number of Classes	Object Classes	Number of Images	Number of Instances
Precast Component Detection	Precast (PC)	6	beam, column, person, slab, trailer, upright	3771	40,699
Worker and PPE Detection	SafetyGear (SG)	3	hardhat, vest, worker	5201	17,304

The PC dataset [40] comprises 3771 images containing 40,699 annotated instances across six object classes: beam, column, person, slab, trailer, and upright. This dataset targets automated precast concrete component identification in construction environments, characterized by high instance density (10.79 instances per image) and significant class imbalance, with the column class representing 72.2% of total instances.

The SG dataset [41] contains 5201 images with 17,304 annotated instances across three classes: hardhat, vest, and worker. This dataset targets worker and PPE object detection

in construction sites, exhibiting moderate instance density (3.33 instances per image) and relatively balanced class distribution compared to the PC dataset.

2.3.1. Dataset Configuration and Preprocessing

Both datasets underwent preprocessing to ensure compatibility with the detection framework. For the PC dataset, auto-orientation was applied to correct image orientation metadata, followed by resizing to fit within 640 × 640 pixels while maintaining aspect ratios. Data augmentation was implemented during training with six output variations per example, including brightness adjustments (between −25% and 0%) and noise injection (up to 1.99% of pixels). These augmentations were applied to address the dataset's limited scale and to improve model robustness to lighting variations.

The SG dataset preprocessing included auto-orientation and resizing with stretching to 640 × 640 pixels. Class modifications were performed to consolidate the original annotation scheme, with seven classes remapped and two classes dropped to create the final three-class configuration. No augmentations were applied to this dataset, maintaining the original data distribution for evaluation.

2.3.2. Train–Validation–Test Split

Table 2 presents the detailed instance distribution and split configuration for both datasets. The split ratios differ between datasets to accommodate their distinct characteristics. The PC dataset employs an 84.9/12.8/2.4 split (3201/481/89 images), allocating a larger proportion to training to accommodate the following: (a) higher instance density and diverse training scenarios; (b) severe class imbalance where minority classes require sufficient examples; (c) the complexity of precast component detection in cluttered construction scenes. The PC test set contains 907 instances across six classes. Two classes (slab and trailer) have zero test instances due to their limited availability in the source dataset.

The PC dataset's highly imbalanced class distribution necessitates split-specific evaluation strategies. Test-set evaluation is employed for adequately represented classes (beam, column, person, upright), while validation set evaluation is used for underrepresented classes (slab, trailer) that lack test instances due to constraints of the source dataset [40]. This approach ensures comprehensive performance assessment while maintaining methodological transparency. However, per-class metrics for these two groups are not directly comparable given the different evaluation splits, and the model rankings based on overall mAP@50 should be interpreted with this limitation in mind.

Table 2. Detailed statistics and split configuration for PC and SG datasets.

Dataset	Class	Train	Val	Test	Total	%
PC	Beam	2883	407	143	3433	8.4
	Column	25,135	3742	506	29,383	72.2
	Person	4688	710	176	5574	13.7
	Slab	271	22	0 *	293	0.7
	Trailer	485	79	0 *	564	1.4
	Upright	1157	213	82	1452	3.6
	Total Instances	34,619	5173	907	40,699	100.0
	Total Images	3201	481	89	3771	—
	Split Ratio (%)	84.9	12.8	2.4	100.0	—
	Avg Inst./Image	10.82	10.75	10.19	10.79	—
SG	Hardhat	5967	1641	878	8486	49.0
	Vest	4569	1311	659	6539	37.8
	Worker	1641	431	207	2279	13.2
	Total Instances	12,177	3383	1744	17,304	100.0

Total Images	3658	1033	510	5201	—
Split Ratio (%)	70.3	19.9	9.8	100.0	—
Avg Inst./Image	3.33	3.27	3.42	3.33	—

Note: * Zero instances in test set due to limited availability in the source dataset.

The SG dataset follows a 70.3/19.9/9.8 split (3658/1033/510 images), which aligns with common practice given its larger scale and more balanced class distribution. The test set comprises 1744 instances with complete coverage across all three classes, enabling evaluation across the full range of worker and PPE detection scenarios.

2.4. Experimental Setup

The experimental environment was configured to ensure reproducible and efficient model training and evaluation. All computations were performed on a high-performance workstation featuring an Intel Xeon W-2265 CPU, an NVIDIA RTX 3090 GPU with 24GB VRAM, and 64GB of system memory, under the Windows 11 operating system. The software stack included Python 3.9.7, PyTorch 2.2.2 as the core deep learning framework and CUDA 11.8 for GPU acceleration. Essential libraries such as NumPy 1.24.3, OpenCV 4.8.0, and Matplotlib 3.7.1 were employed for data manipulation, image processing, and visualization.

All model variants were trained using identical hyperparameters (Table 3) to ensure fair performance comparison. The training configuration adopts established YOLOv11 protocols from the Ultralytics framework, with key parameters selected to accommodate the specific characteristics of construction monitoring datasets.

Table 3. Training configuration for all model variants.

Category	**Parameter**	**Value**
Optimization	Optimizer	AdamW (auto)
	Initial Learning Rate (lr0)	0.01
	Final LR Ratio (lrf)	0.01
	LR Schedule	Linear decay
	Momentum	0.937
	Weight Decay	0.0005
	Warmup Epochs	3
	Warmup Momentum	0.8
	Warmup Bias LR	0.1
Training Protocol	Total Epochs	200
	Early Stopping Patience	100 epochs
	Batch Size	16
	Input Resolution	640 × 640 pixels
	Mixed Precision (AMP)	Enabled
	Pretrained Initialization	COCO weights
	Close Mosaic	Last 10 epochs

The optimizer was configured as 'auto' in the Ultralytics framework, which defaults to AdamW for object detection tasks, selected for its adaptive learning rate properties and compatibility with mixed-precision training. A training duration of 200 epochs was used to facilitate sufficient learning given the dataset characteristics, while early stopping with 100-epoch patience prevents overfitting. All models were initialized with COCO-pretrained weights to leverage transfer learning benefits. The batch size of 16 represents the maximum stable configuration within 24GB VRAM constraints for 640 × 640 resolution training.

A single, fixed set of hyperparameters (Table 3) was used for all model variants, following the default YOLOv11 training protocol established in the Ultralytics framework. Crucially, no architecture-specific hyperparameter tuning or optimization was conducted. This deliberate, controlled approach ensures that any observed performance differences can be attributed directly to the architectural modifications in the neck, rather than to variations in optimization schedules or regularization policies. The use of identical settings across all variants establishes a fair baseline for comparative evaluation, a common practice in architectural comparison studies [42].

2.5. Evaluation Methodology

Model performance was assessed using established object detection metrics aligned with the COCO evaluation protocol [43]. These metrics collectively measure both localization precision and classification accuracy.

The fundamental measure for localization is the intersection over union (IoU), which quantifies the overlap between a predicted bounding box and its corresponding ground truth. Detection accuracy is further characterized by Precision—the ratio of true positive detections to all positive predictions—and Recall—the ratio of true positives to all actual ground truth objects. A detection is considered a true positive if its IoU with a ground truth box exceeds a predefined threshold (typically 0.5).

To provide a comprehensive summary of performance, average precision (AP) is used, which calculates the area under the precision–recall curve for a single object class. This is extended to the multi-class scenario via the mAP, which averages the AP values across all classes. Two standard variants are employed: mAP@50, computed at an IoU threshold of 0.5, and the more stringent mAP@50–95, which represents the average mAP across IoU thresholds from 0.5 to 0.95 in 0.05 increments, emphasizing precise localization.

The training process optimized a composite loss function, L_{total}, composed of three key components: a bounding box regression loss (L_{box}), a classification loss (L_{cls}), and a distribution focal loss (L_{dfl}) for refined box probability estimation. The overall loss is defined as (Equation (1)):

$$L_{total} = \lambda_{box} L_{box} + \lambda_{cls} L_{cls} + \lambda_{dfl} L_{dfl} \quad (1)$$

where the weighting coefficients ($\lambda_{box} = 7.5$, $\lambda_{cls} = 0.5$, $\lambda_{dfl} = 1.5$) follow the default configuration of the YOLOv11 architecture. The relatively high weight assigned to bounding box loss emphasizes the importance of precise spatial localization in object detection.

3. Results

3.1. Training Convergence

3.1.1. PC Dataset

Figure 1 presents the training and validation loss convergence for all evaluated variants on the PC dataset over 200 epochs. All models converged rapidly, stabilizing within 50 epochs. Final training box losses were low (0.50–0.52) with negligible training–validation gaps (±0.01), indicating excellent generalization. MSDC-YOLO and DAM-YOLO achieved the lowest final box losses.

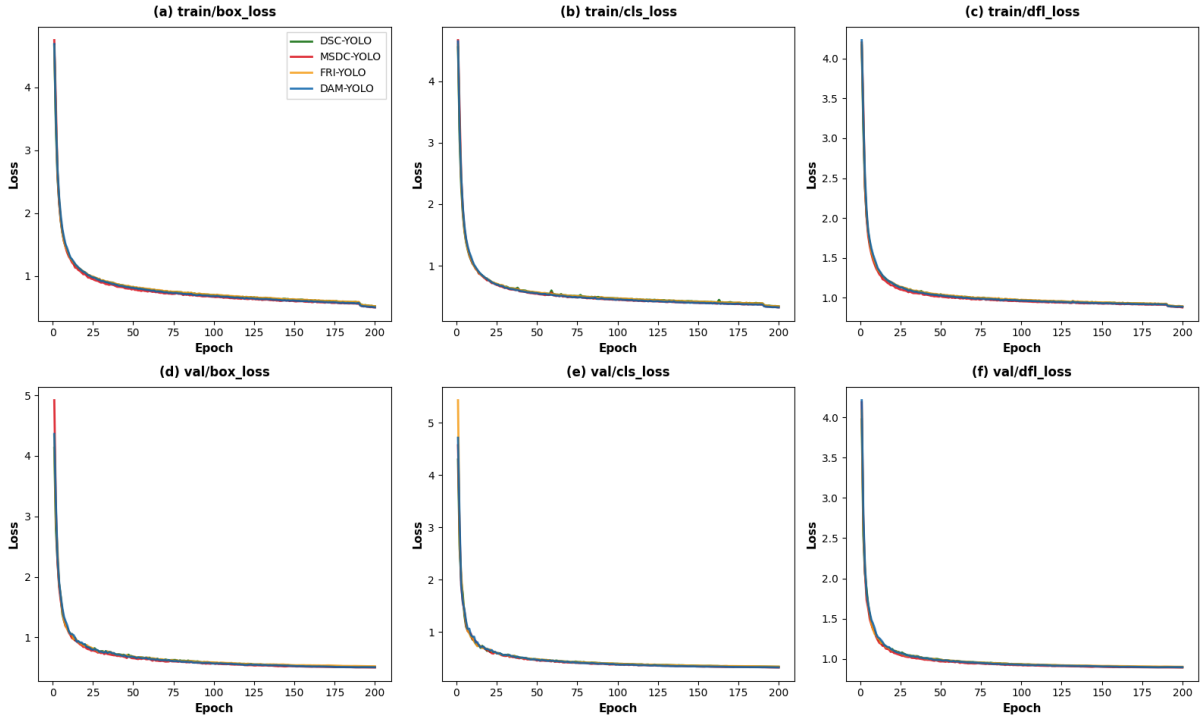


Figure 1. Training and validation loss convergence on PC dataset.

Classification losses exhibited similar convergence patterns across all variants, with narrow final ranges (≤0.02 difference) indicating comparable classification performance regardless of architectural modifications. The steep initial descent phase (epochs 1–50) was followed by gradual refinement, demonstrating stable training dynamics. Distribution focal losses showed consistent behavior across architectures, with all models achieving comparable final performance levels and maintaining stable convergence patterns.

3.1.2. SG Dataset

Figure 2 presents the training and validation loss convergence for all evaluated variants on the SG dataset. Convergence on the SG dataset presented a greater challenge. Although DAM-YOLO and MSDC-YOLO again led in minimizing training box loss, all models exhibited larger training–validation gaps (0.13–0.16), suggesting more difficulty in generalizing to unseen safety gear data compared to precast components.

Classification losses demonstrated DAM-YOLO leading performance, with all architectures showing rapid initial convergence followed by steady improvement. Training–validation gaps remained consistent across variants, suggesting comparable classification generalization capabilities. Distribution focal losses exhibited similar patterns, with final values reflecting the increased complexity of worker and PPE detection compared to precast component identification.

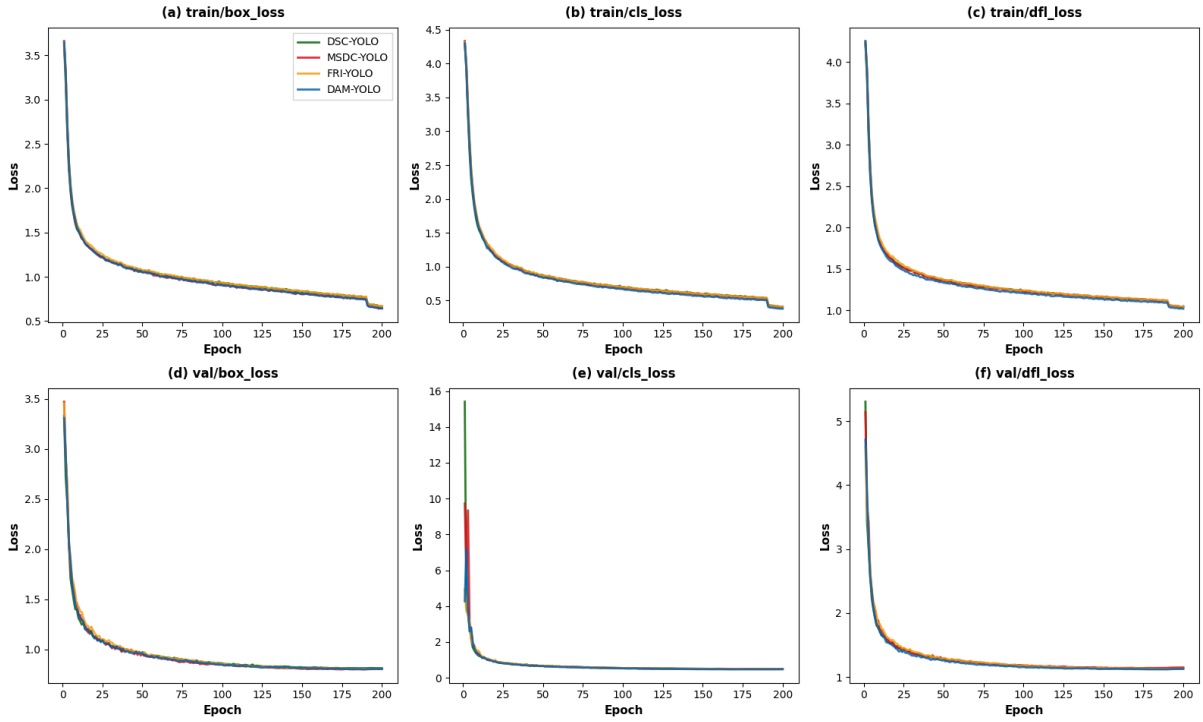


Figure 2. Training and validation loss convergence on SG dataset.

3.2. Detection Performance

3.2.1. PC Dataset

Figure 3 illustrates the detection performance evolution throughout training on the PC dataset. Precision metrics achieved final values spanning 91.56–93.90%, with FRI-YOLO and DSC-YOLO leading performance, while all models demonstrated rapid initial improvement within 50 epochs followed by gradual refinement.

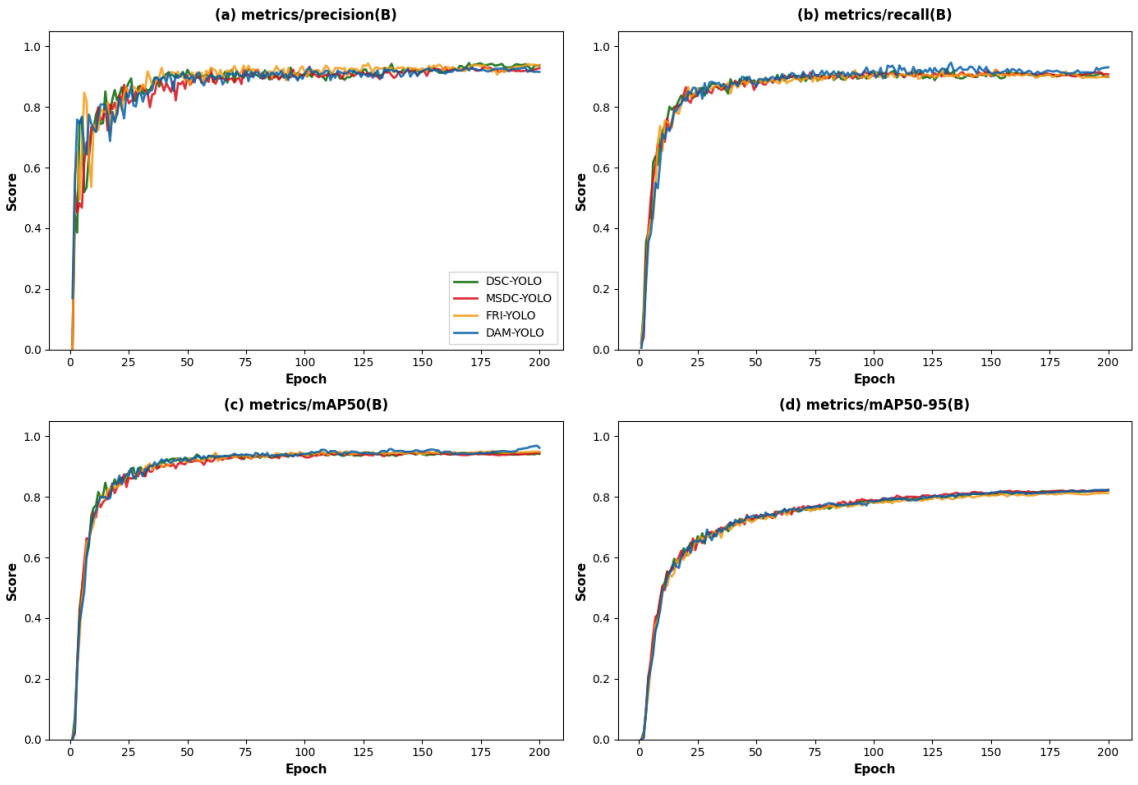


Figure 3. Detection performance metrics on PC dataset.

DAM-YOLO achieved the highest mAP@50 (0.972), representing a 3.0 percentage point improvement over the next best variant. Performance gaps narrowed under the stricter mAP@50–95 metric (1.1 percentage point spread), revealing similar localization precision across all models.

3.2.2. SG Dataset

Figure 4 illustrates detection performance metrics throughout training on the SG dataset. Precision values achieved a narrow final range (0.66 percentage points), with MSDC-YOLO and DAM-YOLO leading the performance, indicating similar discriminative capabilities across the architectures. All models demonstrated rapid convergence within 50 epochs.

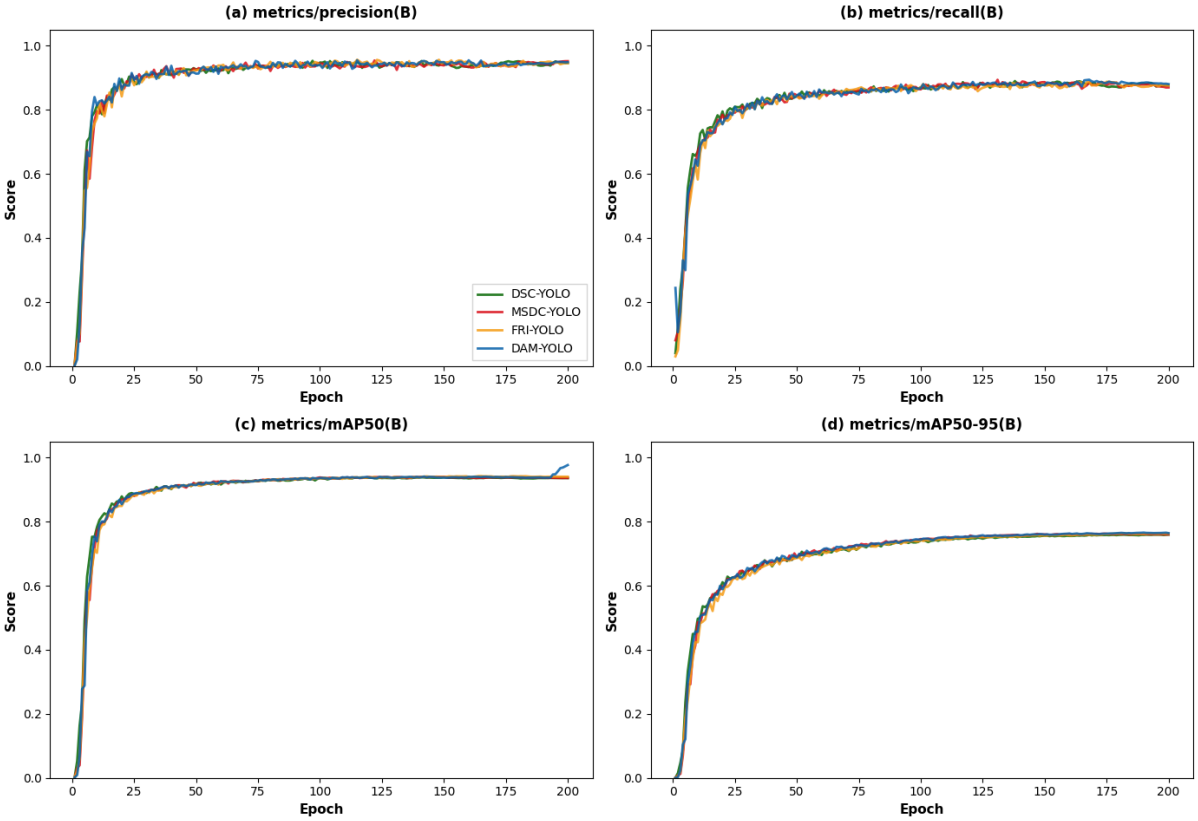


Figure 4. Detection performance metrics on SG dataset.

Recall performance showed reduced variation compared to the PC dataset (1.0 percentage point spread), with DAM-YOLO achieving the highest detection coverage. The smaller performance spread suggests more consistent architectural behavior on worker and PPE detection tasks.

mAP@50 evaluation revealed DAM-YOLO achieving substantial performance advantages (3.2 percentage point lead), while under the more stringent mAP@50–95 metric, performance differences compressed significantly (0.5 percentage point spread), indicating comparable localization accuracy across variants.

3.3. Per-Class Performance Analysis

3.3.1. PC Dataset

Table 4 presents detailed per-class detection performance evaluated on the PC dataset using different splits due to class-specific data availability. Overall performance

patterns revealed precision ranging 0.916–0.939, recall spanning 0.898–0.931, with mAP@50 values between 0.942–0.972 and mAP@50–95 maintaining a narrow band of 0.813–0.825.

Table 4. Per-class metrics comparison on PC dataset. Results for beam, column, person, and upright are evaluated on the test set (†); results for slab and trailer are evaluated on the validation set (‡) due to the absence of test instances for these classes in the source dataset. Direct cross-class comparisons between these two groups should be interpreted with caution given the different evaluation splits.

Model	Class	Box (*P*)	*R*	mAP50	mAP50–95
DSC-YOLO	All	0.937	0.899	0.942	0.821
	Beam †	0.961	0.960	0.988	0.933
	Column †	0.984	0.979	0.993	0.914
	Person †	0.937	0.791	0.929	0.605
	Slab ‡	0.852	0.818	0.837	0.771
	Trailer ‡	0.908	0.872	0.916	0.781
	Upright †	0.981	0.972	0.987	0.921
MSDC-YOLO	All	0.928	0.908	0.944	0.825
	Beam †	0.951	0.958	0.986	0.940
	Column †	0.983	0.979	0.992	0.914
	Person †	0.932	0.829	0.934	0.622
	Slab ‡	0.822	0.818	0.860	0.769
	Trailer ‡	0.897	0.886	0.908	0.778
	Upright †	0.983	0.977	0.986	0.925
FRI-YOLO	All	0.939	0.898	0.949	0.813
	Beam †	0.967	0.953	0.986	0.925
	Column †	0.982	0.983	0.993	0.914
	Person †	0.930	0.830	0.925	0.601
	Slab ‡	0.906	0.818	0.914	0.756
	Trailer ‡	0.863	0.835	0.896	0.770
	Upright †	0.988	0.972	0.979	0.912
DAM-YOLO	All	0.916	0.931	0.972	0.824
	Beam †	0.934	0.980	0.996	0.929
	Column †	0.973	0.988	0.993	0.911
	Person †	0.884	0.893	0.966	0.613
	Slab ‡	0.823	0.846	0.954	0.786
	Trailer ‡	0.888	0.899	0.964	0.786
	Upright †	0.992	0.981	0.988	0.918

Class-level analysis on the PC dataset test set (89 images, 907 instances) revealed consistent patterns across the evaluated variants. Column and beam detection achieved the highest AP50 values across all variants, with DAM-YOLO reporting AP50 of 0.993 and 0.996 for column and beam, respectively. Column represents 72.2% of test instances (506 of 907). Upright detection demonstrated strong performance across all variants (DAM-YOLO: AP50 = 0.988, AP50–95 = 0.918, 82 test instances). Person detection presented the greatest localization challenge, with AP50–95 values constrained to 0.601–0.622 across all evaluated variants—substantially lower than structural classes (AP50–95 range: 0.911–0.940). The beam class achieved R = 0.980 for DAM-YOLO on the 143-instance test set, indicating low missed beam detections within this test partition. The lower precision values for person detection (0.884–0.937 across variants) suggest that false positives may arise from background regions containing scaffolding and equipment at similar scales, though

per-image analysis would be required to confirm this pattern. Slab and trailer classes, which have zero test instances due to constraints of the source dataset [40], are evaluated on the validation set as reported in Table 4 (‡); slab AP50 ranges from 0.837–0.954 and trailer AP50 ranges from 0.896–0.964 across all variants.

3.3.2. SG Dataset

Table 5 summarizes per-class detection metrics for worker and PPE classes on the SG dataset.

Overall performance patterns differed notably from the PC dataset, with precision ranging 0.938–0.952, recall spanning 0.869–0.885, and mAP@50 values between 0.936–0.968.

Table 5. Per-class metrics comparison on SG dataset.

Model	Class	Box (*P*)	*R*	mAP50	mAP50–95
DSC-YOLO	All	0.946	0.876	0.936	0.760
	Hardhat	0.956	0.870	0.942	0.712
	Vest	0.911	0.862	0.937	0.723
	Worker	0.970	0.896	0.929	0.844
MSDC-YOLO	All	0.938	0.885	0.939	0.764
	Hardhat	0.943	0.882	0.947	0.717
	Vest	0.908	0.876	0.938	0.726
	Worker	0.961	0.896	0.931	0.848
FRI-YOLO	All	0.952	0.869	0.942	0.763
	Hardhat	0.958	0.853	0.942	0.711
	Vest	0.931	0.857	0.944	0.733
	Worker	0.969	0.898	0.938	0.845
DAM-YOLO	All	0.945	0.885	0.968	0.766
	Hardhat	0.950	0.879	0.972	0.717
	Vest	0.923	0.870	0.981	0.731
	Worker	0.962	0.905	0.991	0.850

Class-level analysis on the SG dataset test set (510 images, 1744 instances) revealed distinct performance patterns across the three classes. Hardhat detection achieved AP50 of 0.942–0.972 and AP50–95 of 0.711–0.717 across all variants. The lower recall values for hardhat detection suggest that false negatives may be associated with overhead viewing angles where crown geometry is less discriminative and with partial occlusion by equipment, though these patterns require per-image verification. Vest detection achieved AP50 of 0.937–0.981 but AP50–95 of 0.723–0.733—the lowest localization accuracy among SG classes despite constituting 37.8% of dataset instances. The lower localization accuracy for vest detection suggests that false positives may involve background regions with similar color and texture profiles, and false negatives may occur under conditions of color-background similarity or partial concealment, consistent with documented challenges in PPE detection [16]. Worker detection demonstrated the highest AP50–95 values (0.844–0.850) across all variants despite representing only 13.2% of dataset instances (DAM-YOLO test set: AP50 = 0.991, AP50–95 = 0.850, 207 instances). The lower recall for worker detection suggests false negatives may be associated with severe occlusion cases, consistent with documented detection challenges in cluttered environments [20].

The relatively balanced class distribution (49.0% hardhat, 37.8% vest, 13.2% worker) resulted in more uniform performance across classes compared to the highly imbalanced PC dataset, with all models achieving mAP@50–95 values of at least 0.760.

3.4. Comparative Summary

Table 6 presents a comprehensive comparison across both datasets. On the PC dataset, DAM-YOLO achieved the highest mAP@50 (0.972), while MSDC-YOLO demonstrated the highest mAP@50–95 (0.825). DAM-YOLO achieved the highest mAP@50 at 0.972, representing a 3.0 percentage point improvement over the lowest-performing variant (DSC-YOLO at 0.942).

On the SG dataset, overall mAP@50 values exhibited a comparable range, spanning 0.936 (DSC-YOLO) to 0.968 (DAM-YOLO). However, mAP@50–95 values on the SG dataset (0.760–0.766) were consistently lower than those on the PC dataset (0.813–0.825), with an average difference of 5.7 percentage points. Precision on the SG dataset demonstrated higher values (0.938–0.952) compared to the PC dataset (0.916–0.939), while recall patterns showed greater variability (0.869–0.885 on SG versus 0.898–0.931 on PC).

Table 6. Overall performance comparison across datasets with computational profiling metrics.

Model	Dataset	Precision	Recall	mAP@50	mAP@50–95	Params (M)	GFLOPs
DSC-YOLO	PC	0.937	0.899	0.942	0.821	2.591	6.4
	SG	0.946	0.876	0.936	0.760		
MSDC-YOLO	PC	0.928	0.908	0.944	0.825	3.883	8.8
	SG	0.938	0.885	0.939	0.764		
FRI-YOLO	PC	0.939	0.898	0.949	0.813	2.678	6.6
	SG	0.952	0.869	0.942	0.763		
DAM-YOLO	PC	0.916	0.931	0.972	0.824	3.002	7.2
	SG	0.945	0.885	0.968	0.766		

Note: Parameter counts are reported for the PC dataset configuration (6 classes); values for the SG dataset configuration (3 classes) differ by 585–600 parameters due to class-specific detection head output dimensions. GFLOPs computed at 640 × 640 input resolution.

Across both datasets, DAM-YOLO achieved the highest observed mAP@50 in single-run evaluation (0.972 on PC, 0.968 on SG), though the performance spread remained narrow at 3.0 percentage points (PC) and 3.2 percentage points (SG). MSDC-YOLO achieved the highest mAP@50–95 on the PC dataset (0.825), though this did not extend to the SG dataset (0.764). FRI-YOLO achieved the highest precision on the SG dataset (0.952) with moderate mAP@50–95 on both datasets (0.813 on PC, 0.763 on SG). DSC-YOLO demonstrated comparable performance across both datasets (mAP@50: 0.942 on PC, 0.936 on SG). These narrow differences suggest that all four variants yield broadly comparable detection accuracy within this framework.

The absolute performance differences between the best and worst performing models remained relatively narrow on both datasets. On the PC dataset, the mAP@50 spread was 3.0 percentage points (0.942–0.972), while on the SG dataset, it was 3.2 percentage points (0.936–0.968). For the more stringent mAP@50–95 metric, the performance variation was even more constrained, with a range of 1.2 percentage points on the PC dataset (0.813–0.825) and 0.6 percentage points on the SG dataset (0.760–0.766).

Per-class analysis revealed that certain object categories exhibited consistent difficulty across models. On the PC dataset, person detection achieved mAP@50–95 values of only 0.601–0.622 across all models, substantially lower than the 0.911–0.940 range observed for beam and column classes. On the SG dataset, vest detection proved most challenging in terms of precise localization, with all models achieving mAP@50–95 values below 0.733, compared to worker detection which exceeded 0.844 for all variants.

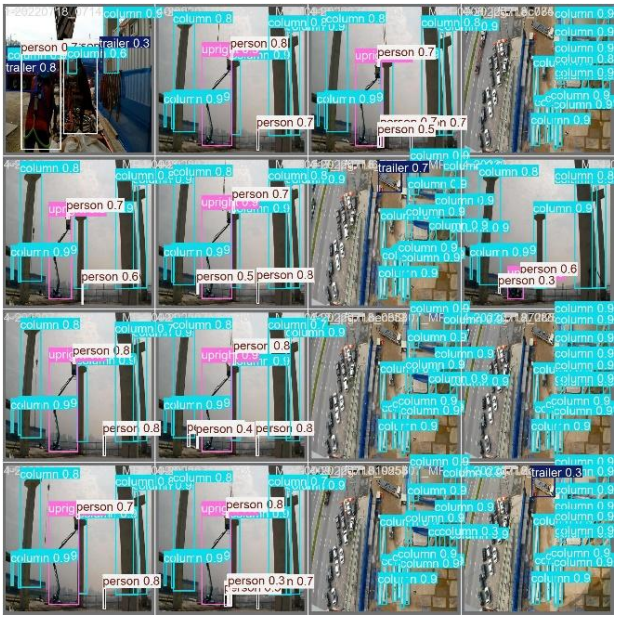
In terms of computational cost, DSC-YOLO achieves the lowest parameter count (2.591M, 6.4 GFLOPs), while MSDC-YOLO incurs the highest cost (3.883M, 8.8 GFLOPs) due to three parallel dilated convolution branches per neck block. FRI-YOLO and DAM-

YOLO occupy intermediate positions at 2.678M (6.6 GFLOPs) and 3.002M (7.2 GFLOPs), respectively. These differences should be interpreted alongside the detection accuracy results: the variant with the lowest parameter count (DSC-YOLO) does not achieve the highest mAP@50, and the variant with the highest computational cost (MSDC-YOLO) does not consistently outperform others, indicating that parameter count alone is not a reliable predictor of detection performance within this framework.

3.5. Visual Detection Performance

Figure 5 presents visual detection results across diverse precast construction environments, showing multi-class object recognition performance in both indoor construction facilities and outdoor precast concrete storage areas.

Precast elements, particularly columns and uprights, show generally high detection confidence across all methods, reflecting the distinctive geometric characteristics of these construction components. Person and trailer detection exhibit broader performance variations, reflecting the challenges associated with detecting objects with variable appearances, dynamic poses, and potential occlusions in construction environments.

(**a**) DSC-YOLO	(**b**) MSDC-YOLO
	
(**c**) FRI-YOLO	(**d**) DAM-YOLO

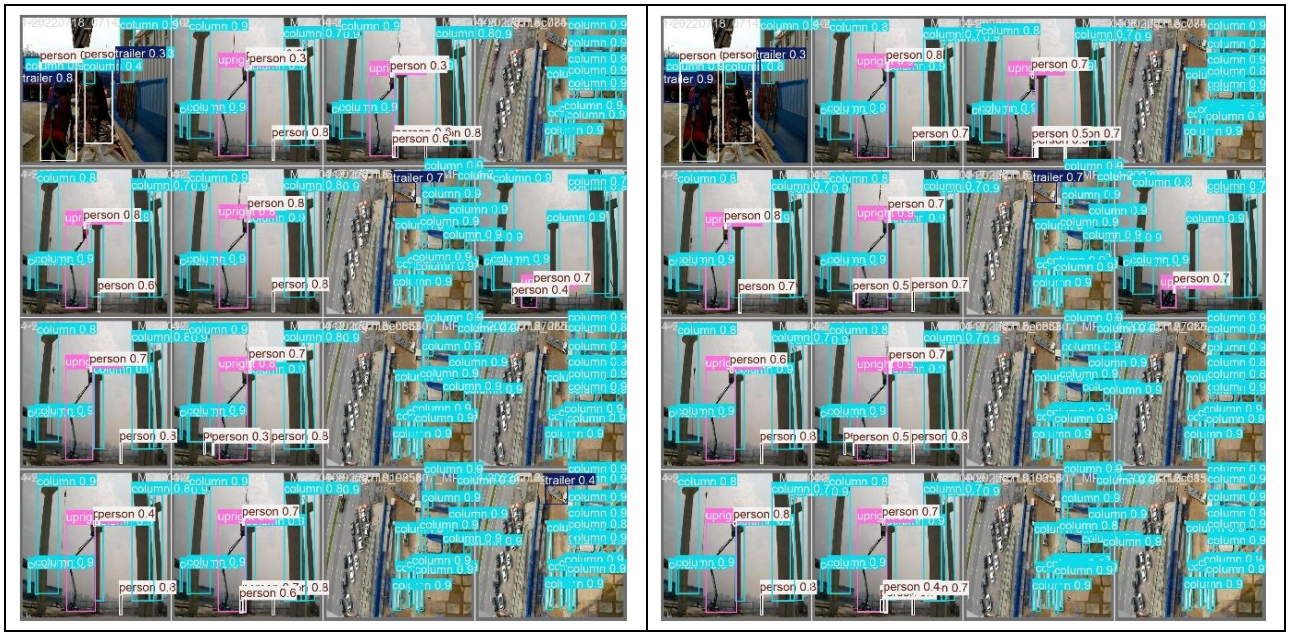
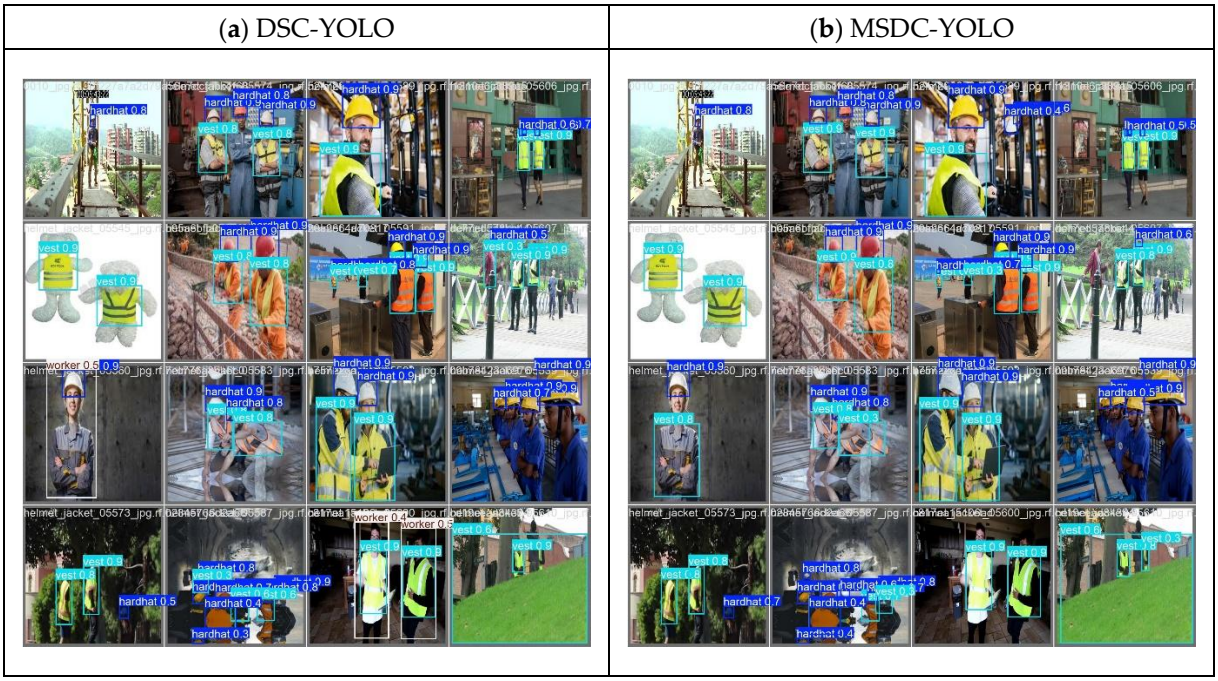


Figure 5. Visual detection results comparison on PC dataset showing (**a**) DSC-YOLO, (**b**) MSDC-YOLO, (**c**) FRI-YOLO, and (**d**) DAM-YOLO performance across diverse construction scenes. Bounding box colors indicate object classes: blue — trailer; cyan — column; white — person; pink — upright. Confidence scores range from 0.3 to 1.0.

Figure 6 demonstrates safety gear detection performance across diverse construction environments, encompassing both indoor confined spaces and outdoor work areas with varied monitoring scenarios. The evaluation includes close-up individual worker detection and wide-angle group surveillance applications.

The results show detection capabilities across all evaluated methods for primary worker and PPE categories. Hardhat and vest detection maintain performance with confidence scores typically spanning the evaluation range, while worker detection shows more variable performance across methods, reflecting the challenges of human detection in dynamic construction environments with varying poses, occlusions, and lighting conditions.



https://doi.org/10.3390/buildings16091728

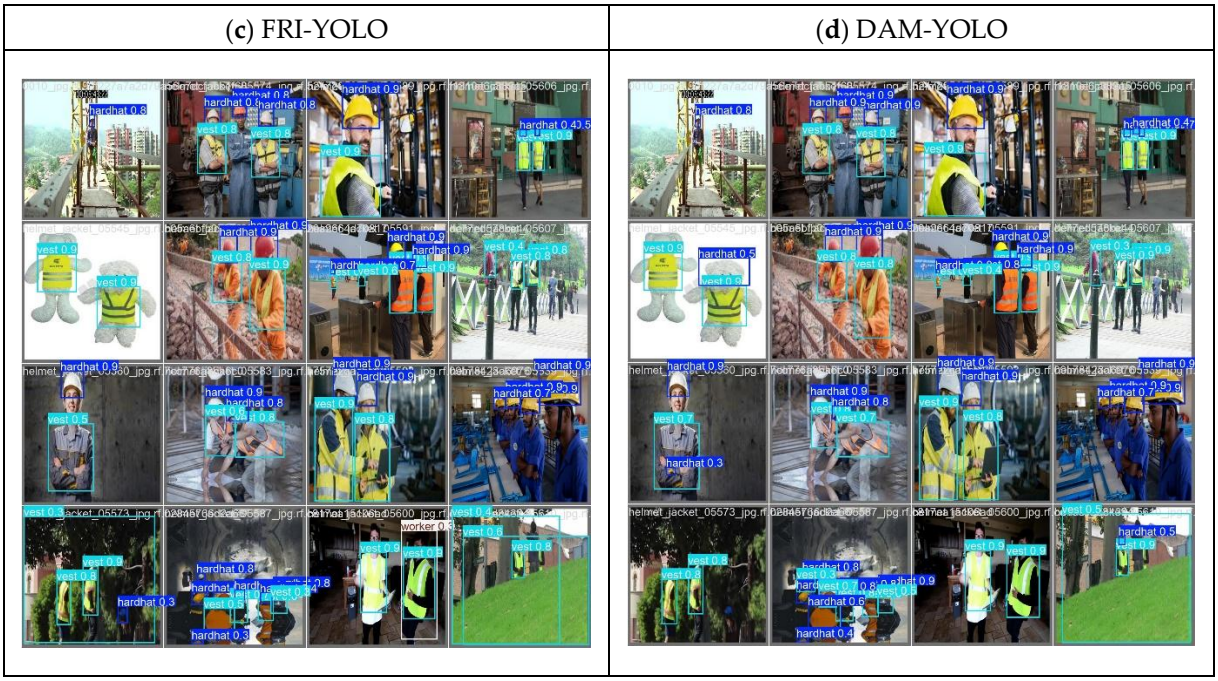


Figure 6. Visual detection results comparison on SG dataset showing (**a**) DSC-YOLO, (**b**) MSDC-YOLO, (**c**) FRI-YOLO, and (**d**) DAM-YOLO performance across diverse construction safety scenarios. Bounding box colors indicate object classes: blue — hardhat; cyan — vest; white — worker. Confidence scores range from 0.3 to 0.9.

4. Discussion

4.1. Interpretation of Main Findings

4.1.1. Performance Comparison

A narrow performance variation was observed across all four evaluated variants (within 3.2 percentage points for mAP@50), suggesting that the neck-level modifications evaluated in this study provide incremental rather than transformative performance differences within the YOLOv11n framework. DAM-YOLO achieved the highest observed mAP@50 on both datasets in single-run evaluation; however, given the narrow absolute differences and the absence of multi-run validation, this ranking should not be interpreted as evidence of architectural superiority.

Distinct precision–recall patterns were quantified: DAM-YOLO favored recall (0.931 recall vs. 0.916 precision on the PC dataset), whereas FRI-YOLO prioritized precision (0.939 precision vs. 0.898 recall on PC; 0.952 precision on SG), reflecting different optimization characteristics inherent to each architectural modification.

4.1.2. Impact of Class Imbalance and Challenging Detection Scenarios

Per-class analysis indicated that detection performance is influenced more by intrinsic object characteristics and environmental challenges than by class frequency alone, although severe imbalance introduces distinct evaluation difficulties. Despite the severe imbalance in the PC dataset, performance on the dominant ‘column’ class remained high across all models (mAP@50–95 ≥ 0.911). This suggests that the training protocol is resilient to skewed distributions for visually distinct, high-frequency objects.

Conversely, evaluating minority classes with zero test instances—trailer (564 total instances, 1.4%) and slab (293 instances, 0.7%)—required validation set assessment, highlighting the practical data collection challenges for rare on-site objects, while the dominant column class comprised 72.2% of instances. The training process showed resilience to such imbalance through the focal loss component, which down-weights well-classified

examples and focuses learning on hard negatives, preventing the dominant classes from overwhelming the minority class gradients. The 200-epoch training schedule and the combined loss function (bounding box, classification, and distribution focal loss) contributed to stable convergence across all models, indicating resilience to imbalance during optimization.

On the SG dataset, performance was not correlated with frequency. The less frequent worker class (13.2% of instances) achieved higher mAP@50–95 (0.844–0.850) than the more prevalent vest class (37.8%, mAP@50–95 of 0.723–0.733). This inverse relationship suggests that the primary detection difficulty for the vest class stems from intrinsic characteristics—such as its deformable shape and color similarity to backgrounds—rather than scarcity. This suggests that vest, despite being a prevalent class in the SG dataset, may face detection challenges less related to sample scarcity than to visual complexity, though controlled analysis would be required to confirm this relationship. For such challenging classes, architectural enhancements or targeted data augmentation for occlusion and deformation may be more effective than simply increasing data volume.

The lower mAP@50–95 scores for person/worker and vest detection align with documented visual challenges in construction environments, including frequent occlusion, deformation, complex backgrounds, and small-scale appearances in wide-angle imagery [16,20]. In summary, while class imbalance necessitates careful evaluation for rare classes, the primary bottleneck for others is their inherent detectability, shifting the focus from dataset balancing to addressing class-specific visual challenges.

4.2. Architectural Insights and Mechanism Analysis

The four modifications target distinct aspects of feature processing within the neck as follows.

- DSC-YOLO: Employs depthwise separable convolutions, achieving a marginal parameter reduction relative to the YOLOv11n baseline while maintaining comparable detection accuracy. This suggests that effective feature extraction is achievable with minimal architectural overhead.
- MSDC-YOLO: Utilizes multi-scale dilated convolutions to expand receptive fields. It yielded the highest mAP@50–95 (0.825) on the PC dataset, suggesting a particular advantage for detecting large, rigid structures.
- FRI-YOLO: Incorporates a lightweight 1 × 1 convolutional refinement interface. It achieved the highest precision (0.952) on the SG dataset, indicating effectiveness in suppressing false positives.
- DAM-YOLO: Integrates dual attention mechanisms (ECA + SE) that allow the model to re-weight channel-wise feature responses. It provided the most consistent top mAP@50 across both construction object detection scenarios in single-run evaluation, consistent with robust feature refinement.

The relatively uniform overall performance indicates that while each strategy optimizes a different aspect, their net effect on detection capability within this framework is comparable. Therefore, selection can be guided by secondary factors like specific precision-recall requirements or integration ease.

4.3. Cross-Dataset Analysis

Performance patterns across the PC and SG datasets elucidate how architectural modifications respond to different detection scenarios. The consistently higher mAP@50–95 on the PC dataset reflects fundamental task differences.

The higher instance density in the PC dataset (10.79 vs. 3.33 per image) may provide more examples for bounding box regression. Furthermore, the rigid geometry and well-

defined boundaries of precast components likely facilitate more precise localization compared to the deformable PPE items in the SG dataset.

Class imbalance influenced models differently. Severe imbalance in the PC dataset did not hinder high performance on the dominant class, whereas the more balanced SG dataset showed greater per-class performance variation. The lower accuracy for the frequent vest class compared to the less frequent worker class reinforces that detection difficulty is often tied to visual characteristics and not frequency.

Generalization patterns also differed. The minimal training–validation gap for box loss on the PC dataset (−0.009 to 0.005) suggests controlled generalization. The larger gap on the SG dataset (0.1351–0.1602) may indicate overfitting or greater distributional shifts within its data, possibly due to higher variability in camera angles, lighting, and site conditions. Despite these differences, all models exhibited similar convergence patterns—rapid improvement in the first 50 epochs followed by gradual refinement—indicating that the architectural modifications operate within a similar optimization landscape.

4.4. Benchmarking Against State-of-the-Art

To contextualize the performance, a comparison with selected state-of-the-art studies in construction object detection is presented in Table 7. Benchmarks were chosen based on recency, high reported accuracy, or direct task relevance.

Table 7. Comparative benchmarking with state-of-the-art construction monitoring studies.

Study	Architecture	Task Domain	Dataset Size (Images)	mAP@50 (%)	Key Innovation/Focus
State-of-the-Art (Comparative Works)					
[22]	YOLOv5 + CSPDarkNet53	Precast Components	12,000	97.0	First dedicated PC dataset
[10]	MARA-YOLO (YOLOv8s)	Multi-PPE Detection	2750	74.7	Comprehensive PPE + non-compliance
[12]	YOLOv5s + CMNet	Helmet + Vest Compliance	8890	93.6	Simultaneous detection; handles imbalance
[44]	DETR (Transformer)	Work-at-heights PPE	Compiled	98.2	Transformer-based; highest reported accuracy
[15]	YOLOv10 + Transformer	Safety Helmet Monitoring	10,356	94.24	Dual-camera strategy; Transformer backbones
This Study (Evaluated Variants)					
DAM-YOLO	YOLOv11n + Dual Attention	PC Detection	3771	97.2	Dual attention mechanisms
		PPE Detection	5201	96.8	
DSC-YOLO	YOLOv11n + Depthwise Sep	PC Detection	3771	94.2	Computational efficiency
		PPE Detection	5201	93.6	

Table 7 highlights the following three aspects of this work.

1. Competitive Performance: The evaluated variants achieve mAP@50 scores competitive with recent studies. It is noted that direct numerical ranking is not definitive due to differences in datasets, class compositions, and evaluation environments. This comparison positions the work within the broader performance landscape of construction vision models.
2. Cross-Scenario Consistency: The architectures maintain high performance across both precast component (up to 97.2% mAP@50) and worker and PPE (up to 96.8% mAP@50) detection scenarios.

3. Systematic Architectural Analysis: The primary contribution is the controlled, empirical comparison of four distinct neck-level modifications under consistent protocols, providing insights into their relative effectiveness for construction monitoring tasks.

4.5. Limitations and Future Directions

Several limitations are acknowledged. First, the architectural exploration was confined to the neck-level; interactions with different backbones or detection heads were not examined. Second, a fixed training configuration was applied uniformly across all variants without variant-specific hyperparameter tuning. As a result, observed performance rankings may reflect differential sensitivity to the default training protocol rather than reflecting inherent architectural superiority; variant-specific tuning may alter relative rankings. Third, the PC dataset test split contains only 89 images (907 instances) across four classes, with slab and trailer absent from the test set due to constraints of the source dataset. This limits the reliability of per-class performance estimates for these two classes and precludes direct cross-class comparisons across all six categories; results for slab and trailer should therefore be interpreted as indicative rather than definitive. Fourth, while parameter counts and GFLOPs for all evaluated variants were obtained in this study, inference speed and memory consumption on target hardware were not measured; these metrics are necessary to fully substantiate deployment recommendations. Additionally, each variant was evaluated using a single training run due to computational resource constraints. The narrow performance differences observed—particularly for mAP@50–95 (0.6 pp on the SG dataset, 1.2 pp on the PC dataset)—may not be statistically significant, and multi-run evaluation with mean ± standard deviation reporting is identified as a priority for future work to establish more robust variant rankings. Fifth, the findings are based on two construction-specific benchmark datasets; cross-site and cross-domain evaluation is required to assess broader generalizability.

Future work should concern the following: (1) evaluate the modifications with different backbones and on larger, more diverse datasets; (2) conduct on-device profiling (e.g., FPS, memory) to quantify parameter-accuracy trade-offs for deployment scenarios; (3) investigate hybrid architectures combining the most effective mechanisms; (4) validate performance in real-time, on-edge deployment scenarios.

5. Engineering Applications

5.1. Implications for Practical Implementation

The systematic comparison provides empirical evidence for architecture selection within the evaluated scenarios. The comparable detection accuracy across all four evaluated variants suggests that practitioners are not constrained to a single architectural choice on accuracy grounds alone, and that secondary factors—such as available computational resources, integration requirements, and precision-recall priorities—may guide final selection. However, the present analysis is limited to detection accuracy and parameter-level metrics; inference latency and memory consumption on target hardware were not measured.

5.2. Deployment Decision Framework

The minimal performance differences suggest that if all variants meet the required accuracy threshold, the selection can prioritize secondary factors such as computational budget or integration requirements. The following observations are based solely on accuracy metrics under controlled benchmark conditions and should not be interpreted as deployment prescriptions without supporting inference latency and memory measurements on target hardware: DAM-YOLO may be better suited to applications where minimizing

missed detections is prioritized, given its highest observed recall across both datasets (PC: 0.931, SG: 0.885); FRI-YOLO may be preferable where false positive suppression is required, given its highest observed precision (PC: 0.939, SG: 0.952). These observations require validation through on-device profiling and real-site deployment before operational adoption.

The DSC-YOLO variant employs depthwise separable convolutions, achieving a marginal parameter reduction relative to the YOLOv11n baseline while maintaining comparable detection accuracy. However, on-device profiling—including inference latency and memory measurements—would be required to fully quantify parameter-accuracy trade-offs for deployment decisions.

The variants maintain standard interfaces, facilitating integration. The analysis of challenging detection conditions suggests that camera placement optimization to mitigate occlusion and viewpoint variability may complement algorithmic improvements, though this inference is drawn from benchmark results rather than field deployment evidence.

5.3. Contribution to Systematic Engineering Evaluation

The primary contribution of this study is methodological rather than architectural: a controlled, empirical comparison of four established neck-level strategies within a unified YOLOv11n framework, evaluated consistently across two construction object detection scenarios. The value of this contribution is threefold as shown below.

1. Controlled Architectural Comparison: A direct comparison of four neck-level strategies (DSC, MSDC, FRI, DAM) under identical training conditions, enabling performance differences to be attributed to architectural choices rather than experimental variability.
2. Cross-Scenario Validation: Consistent evaluation across precast component detection and PPE-related object detection scenarios, demonstrating that conclusions hold across distinct task characteristics within the evaluated benchmark datasets.
3. Empirical Selection Guidance: Translating benchmark results into practitioner-oriented insights, acknowledging that final deployment selection requires additional profiling beyond the parameter counts and GFLOPs that are reported here—particularly inference latency and memory measurements on target hardware.

6. Conclusions

This study presents a controlled empirical comparison of four neck-level architectural modifications within the YOLOv11n framework, evaluated systematically across two construction object detection scenarios. While existing construction monitoring research has predominantly employed YOLOv5, YOLOv8, or YOLOv10 architectures for individual detection tasks, this study assessed four neck-level enhancement strategies—DSC-YOLO, MSDC-YOLO, FRI-YOLO, and DAM-YOLO—across both precast concrete detection and worker and PPE detection scenarios.

The experimental evaluation on two specialized construction datasets—precast component detection (3771 images, 6 classes) and worker and PPE detection (5201 images, 3 classes)—yielded the following three principal findings.

1. High Baseline Robustness: The core YOLOv11n framework demonstrated robust performance, with all variants achieving mAP@50 scores between 0.936 and 0.972, confirming its suitability for diverse construction vision tasks.
2. Incremental Architectural Benefits: The modifications provided refinements rather than transformative gains. DAM-YOLO, incorporating dual attention mechanisms, achieved marginal advantages in overall mAP@50.

3. Task-Dependent Performance Determinants: Detection difficulty appeared more strongly associated with intrinsic object characteristics (e.g., rigidity, deformability, background similarity) than with class frequency, though this observation is based on correlational evidence and warrants further controlled investigation. This suggests that future improvements may benefit more from targeted data strategies and training optimizations than from isolated architectural adjustments.

For practitioners, this analysis guides architecture selection for construction monitoring. Minimal performance variation across all four variants means deployment can prioritize application needs over accuracy alone. Based on single-run benchmark evaluation, DAM-YOLO demonstrated the highest detection coverage, MSDC-YOLO the highest localization precision, FRI-YOLO the lowest false positive rate, and DSC-YOLO the lowest parameter count among the evaluated variants. This flexibility accommodates computational constraints and operational priorities. Future work should focus on dataset quality, occlusion challenges, and domain-adaptive training for diverse construction environments.

This work provides a controlled, empirical comparison of neck-level design trade-offs for construction object detection across two benchmark scenarios. Future work should evaluate these variants in diverse construction environments and validate performance in real-time on-site deployments.

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Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the author.

Conflicts of Interest: The author declares no conflicts of interest.

Appendix A

Table A1. Representative studies on PPE detection and precast component identification.

Study	PPE/Component Focus	Base Architecture	Key Technical Innovations	Object Classes	Dataset (Size)	mAP@50 (%)	Notable Contribution	Limitations
[22]	Precast concrete components	YOLOv5 + CSPDarkNet53	Systematic dataset development; automated document preparation	beam, column, slab, upright, trailer, person	12,000 images	97.0	First multi-class PC dataset with field validation	No logistics elements; occlusion degradation
[10]	Multi-class PPE + negatives	MARA-YOLO (YOLOv8s)	AS-Block downsampling; R-C2F dilated convolutions; relational PPE modeling	hardhat, mask, no-head-PPE, gloves, No-gloves, safety vest, no-safety vest, no-PPEs, safety cone	2750 images, 17,379 entities	74.7	Most comprehensive PPE system with negative detection	Small object challenges; lighting sensitivity

[12]	Helmet + vest compliance	YOLOv5s + CMNet + GSANeck	CNN-MHSA hybrid; GSA attention; 18% parameter reduction	reflective vest, no reflective vest, hat, person	8890 images	93.6	First simultaneous helmet + vest detection addressing imbalance	No harness; no annotation variation handling
[44]	Work-at-heights PPE	DETR (Transformer)	Position-aware MHSA; dual-label assignment	helmet, no-helmet, harness	compiled from diverse sources	98.2	Highest mAP; first DETR for construction PPE	Low speed (15–18 FPS); limited PPE types
[23]	Safety helmet variations	YOLOR	Complete relabeling (3→6 classes); 10+ model benchmark	6 helmet variation classes	5000 images, 75,570 labels	88.3	Demonstrates annotation variation importance	No vest/harness; limited scale
[24]	Colored hardhat taxonomy	YOLOX-m (anchor-free)	Color-coded taxonomy; weather robustness testing	4 hardhat colors, vest, person	1699 images, 11,604 objects	89.8	Comprehensive color taxonomy; weather evaluation	Occlusion issues; weather degradation
[45]	Hard hat terminology	YOLOv5s + MobileNetV3 + CA	Knowledge distillation, Model size: ~39% reduction	helmet, head_with_helmet, face, person_with_helmet	5000 images, 75,578 labels	82.2	Addresses “hardhat” vs “helmet” terminology	Limited to helmet only; power plant specific
[16]	Helmet + reflective clothing	Enhanced YOLOv5 + CA	Semi-automatic labeling; SEIoU loss	hat, reflective-clothes, person, other-clothes	7400 images	91.6	Semi-automatic dataset creation reducing burden	Feature interference; imbalance issues
[46]	Safety harness fall prevention	Efficient-YOLOv5	GhostNet backbone; matching-recheck strategy	person, safety harness, clothing	3300 images	94.0	Edge deployment (Jetson Nano); lightweight (8.95 MB)	Severe occlusion fails (>90% coverage)
[13]	Helmet + vest detection	YOLOv8n + CISNeck + MHSA	Fourth detection head (160 × 160); SFFM attention	helmet, no-helmet, vest, no-vest	4408 + 7581 (SHWD)	91.6, 92.5	Superior small target detection; 114 FPS	Increased overhead; no cross-environment validation
[6]	Rebar counting	Faster R-CNN, YOLOv5/v6	Large-scale comparison of 384 models; Real-time UAV video capability	rebar	728 images (UAV)	94.61	Comprehensive benchmark; 32 FPS video processing	Single-class; Performance sensitive to camera perspective
[7]	Rebar counting dataset	Faster R-CNN	First public labeled UAV dataset; Analysis of 8 augmentations	rebar	874 raw images	~90.54 (avg)	Public dataset release; Systematic augmentation analysis	Limited environmental diversity; Single-task focus

[8]	Rebar counting augmentation	Faster R-CNN, YOLOv10 + Transformer	Systematic evaluation of 10 augmentation methods; Architecture-specific recommendations	rebar	728 images	87.71	Introduced rebar-count accuracy metric; Probabilistic sampling policy	Focus on augmentation optimization; Single-class limitation
[14]	Multi-class non-PPE	YOLOv8 + Transformer (ViT/Swin/PVT)	Systematic evaluation of 8 augmentation techniques for non-PPE detection	Non-helmet, Non-mask, Non-glove, Non-vest, Non-shoes	2302 images	73.12	Task- and class-dependent augmentation effectiveness	Focuses only on missing PPE; Limited geographic diversity
[5]	Automated multi-class non-PPE	YOLOv10 + Transformer (Swin/ViT/PVT)	Evaluation on mixed data sources; Transformer backbone comparison	Non-helmet, Non-mask, Non-vest, Non-glove, Non-shoes	3392 images	87.32	Comprehensive multi-source evaluation	Generalization across unseen datasets needs improvement
[15]	Safety helmet monitoring	YOLOv10 + Transformer (Swin/ViT/PVT)	Dual-camera strategy; CNN vs. Transformer backbone comparison	safety helmet, nonsafety helmet	10,356 total images	94.24 (surveillance), 90.86 (body-worn)	Practical dual-camera deployment analysis; ROI calculation	Binary classification; Reduced accuracy for small objects
[21]	Heavy equipment detection	YOLOv10 + Transformer (Swin/ViT/PVT)	Transformer backbones integrated for equipment detection	loader, dump truck, excavator, Remicon, crane	4732 images	89.72	High accuracy with real-time speed (35 FPS)	Limited to equipment detection; No PPE/worker integration
This study (2026)	Precast components + PPE	YOLOv11n + 4 variants	Neck-level modifications (DSC, MSDC, FRI, DAM)	PC: beam, column, slab, upright, trailer, person; SG: hardhat, vest, worker	PC: 3771; SG: 5201 images	PC: 94.2–97.2; SG: 93.6–96.8	Systematic architectural comparison within YOLOv11n across two construction object detection scenarios	PPE detection limited to 3 core classes

Appendix B

Algorithm A1: YOLOv11n

Input: $I \in \mathbb{R}^{3\times640\times640}$

Output: $D = \{(b_i, c_i, s_i)\}_{i=1}^{N}$

```
# Stem Convolutions
x ← Conv(I, 16, k=3, s=2, p=1)                    // [16, 320, 320]
x ← Conv(x, 32, k=3, s=2, p=1)                    // [32, 160, 160]

# Stage 1
```

6: x ← C3k2(*x*, 64, *n*=1, shortcut=False) // [64, 160, 160]
7: x ← Conv(*x*, 64, *k*=3, *s*=2, *p*=1) // [64, 80, 80]
8:
9: # Stage 2
10: F_4 ← C3k2(*x*, 128, *n*=1, shortcut=False) // [128, 80, 80]
11: x ← Conv(F_4, 128, *k*=3, *s*=2, *p*=1) // [128, 40, 40]
12:
13: # Stage 3
14: F_6 ← C3k2(*x*, 128, *n*=1, shortcut=True) // [128, 40, 40]
15: x ← Conv(F_6, 256, *k*=3, *s*=2, *p*=1) // [256, 20, 20]
16:
17: # Stage 4
18: x ← C3k2(*x*, 256, *n*=1, shortcut=True) // [256, 20, 20]
19: x ← SPPF(*x*, 256, *k*=5) // [256, 20, 20]
20: F_{10} ← C2PSA(*x*, 256, *n*=2, *e*=0.5) // [256, 20, 20]
21:
22: # Top-down Path (FPN)
23: x ← Upsample(F_{10}, scale=2) // [256, 40, 40]
24: x ← Concat(*x*, F_6) // [384, 40, 40]
25: F_{13} ← C3k2(*x*, 128, *n*=2, shortcut=False) // [128, 40, 40]
26:
27: x ← Upsample(F_{13}, scale=2) // [128, 80, 80]
28: x ← Concat(*x*, F_4) // [256, 80, 80]
29: P_3 ← C3k2(*x*, 64, *n*=2, shortcut=False) // [64, 80, 80]
30:
31: # Bottom-up Path (PAN)
32: x ← Conv(P_3, 64, *k*=3, *s*=2, *p*=1) // [64, 40, 40]
33: x ← Concat(*x*, F_{13}) // [192, 40, 40]
34: P_4 ← C3k2(*x*, 128, *n*=2, shortcut=False) // [128, 40, 40]
35:
36: x ← Conv(P_4, 128, *k*=3, *s*=2, *p*=1) // [128, 20, 20]
37: x ← Concat(*x*, F_{10}) // [384, 20, 20]
38: P_5 ← C3k2(*x*, 256, *n*=2, shortcut=True) // [256, 20, 20]
39:
40: # Multi-scale Detection
41: *D* ← Detect([P_3, P_4, P_5])
42: return *D*

Algorithm A2: DSC-YOLO

Input: $I \in \mathbb{R}^{(3\times H\times W)}$, RGB image where $H=W=640$
Output: $D = \{(b_i, c_i, s_i)\}_{i=1}^{N}$

1: # Backbone: Feature Extraction (Layers 0-10)
2: F_0 ← Backbone(I) // [256, 20, 20]

```
3:  F4 ← GetFeature(Backbone, layer=4)                    // [128, 80, 80]
4:  F6 ← GetFeature(Backbone, layer=6)                    // [128, 40, 40]
5:
6:  # Neck: Multi-Scale Feature Fusion with DSC
7:  F11 ← Upsample(F0, scale=2)                           // [256, 40, 40]
8:  F12 ← Concat(F11, F6)                                 // [384, 40, 40]
9:  F13 ← DWTinyBlock(F12, C_out=128)                     // [128, 40, 40]
10:
11: F14 ← Upsample(F13, scale=2)                          // [128, 80, 80]
12: F15 ← Concat(F14, F4)                                 // [256, 80, 80]
13: P3 ← DWTinyBlock(F15, C_out=64)                       // [64, 80, 80]
14:
15: F17 ← Conv(P3, stride=2)                              // [64, 40, 40]
16: F18 ← Concat(F17, F13)                                // [192, 40, 40]
17: P4 ← DWTinyBlock(F18, C_out=128)                      // [128, 40, 40]
18:
19: F20 ← Conv(P4, stride=2)                              // [128, 20, 20]
20: F21 ← Concat(F20, F0)                                 // [384, 20, 20]
21: P5 ← DWTinyBlock(F21, C_out=256)                      // [256, 20, 20]
22:
23: # Detection Head
24: D ← Detect([P3, P4, P5])
25: return D
```

Function: DWTinyBlock(*X*, *C*_out)

Input: $X \in \mathbb{R}^{(C_in \times H \times W)}$

Output: $Y \in \mathbb{R}^{(C_out \times H \times W)}$

```
1:  # Depthwise Convolution
2:  X_dw ← Conv2d(X, C_in, kernel=3, groups=C_in, padding=1)
3:  X_dw ← BatchNorm(X_dw)
4:  X_dw ← SiLU(X_dw)
5:
6:  # Pointwise Convolution
7:  Y ← Conv2d(X_dw, C_out, kernel=1, padding=0)
8:  Y ← BatchNorm(Y)
9:  Y ← SiLU(Y)
10:
11: return Y
```

Algorithm A3: MSDC-YOLO

Input: $I \in \mathbb{R}^{(3 \times H \times W)}$, RGB image where $H=W=640$

Output: $D = \{(b_i, c_i, s_i)\}_\{i=1\}^{N}$

```
# Backbone: Feature Extraction
F0, F4, F6 ← Backbone(I)

# Neck: Multi-Scale Dilated Context Aggregation
F11 ← Upsample(F0, scale=2)
F12 ← Concat(F11, F6)
F13 ← DilatedContextBlock(F12, C_out=128)

F14 ← Upsample(F13, scale=2)
F15 ← Concat(F14, F4)
P3 ← DilatedContextBlock(F15, C_out=64)

F17 ← Conv(P3, stride=2)
F18 ← Concat(F17, F13)
P4 ← DilatedContextBlock(F18, C_out=128)

F20 ← Conv(P4, stride=2)
F21 ← Concat(F20, F0)
P5 ← DilatedContextBlock(F21, C_out=256)

# Detection Head
D ← Detect([P3, P4, P5])
return D
```

Function: DilatedContextBlock(*X*, *C*_out)

```
Input:  X ∈ ℝ^(C_in×H×W)
Output: Y ∈ ℝ^(C_out×H×W)

C_mid ← max(1, C_in // 2)

# Parallel Multi-Scale Dilated Convolutions
F1 ← DilatedConv(X, C_mid, dilation=1, padding=1)
F1 ← SiLU(BatchNorm(F1))

F2 ← DilatedConv(X, C_mid, dilation=2, padding=2)
F2 ← SiLU(BatchNorm(F2))

F3 ← DilatedConv(X, C_mid, dilation=3, padding=3)
F3 ← SiLU(BatchNorm(F3))

# Feature Fusion
F_cat ← Concat([F1, F2, F3])                    // [3×C_mid, H, W]
Y ← Conv2d(F_cat, C_out, kernel=1)
Y ← SiLU(BatchNorm(Y))
```

17:

18: return Y

Algorithm A4: FRI-YOLO

Input: $I \in \mathbb{R}^{(3\times H\times W)}$, RGB image where $H=W=640$

Output: $D = \{(b_i, c_i, s_i)\}_\{i=1\}^N$

1: # Backbone & Neck: Standard YOLOv11n

2: $P_3, P_4, P_5 \leftarrow$ YOLOv11n_Neck(I)

3:

4: # Pre-Head Feature Refinement Interface

5: $P_3' \leftarrow$ LinearLiteBlock(P_3) // [64, 80, 80]

6: $P_4' \leftarrow$ LinearLiteBlock(P_4) // [128, 40, 40]

7: $P_5' \leftarrow$ LinearLiteBlock(P_5) // [256, 20, 20]

8:

9: # Detection Head

10: $D \leftarrow$ Detect([P_3', P_4', P_5'])

11: return D

Function: LinearLiteBlock(X)

Input: $X \in \mathbb{R}^{(C\times H\times W)}$

Output: $Y \in \mathbb{R}^{(C\times H\times W)}$

1: # Channel-wise Linear Transformation

2: $Y \leftarrow$ Conv2d(X, C, kernel=1, padding=0)

3: $Y \leftarrow$ BatchNorm(Y)

4: $Y \leftarrow$ SiLU(Y)

5: return Y

Algorithm A5: DAM-YOLO

Input: $I \in \mathbb{R}^{(3\times H\times W)}$, RGB image where $H=W=640$

Output: $D = \{(b_i, c_i, s_i)\}_\{i=1\}^N$

1: # Backbone: Feature Extraction

2: $F_0, F_4, F_6 \leftarrow$ Backbone(I)

3:

4: # Neck: Dual Attention Feature Enhancement

5: $F_{11} \leftarrow$ Upsample(F_0, scale=2)

6: $F_{12} \leftarrow$ Concat(F_{11}, F_6)

7: $F_{13} \leftarrow$ TAFHybridBlock(F_{12}, C_out=128)

8:

9: $F_{14} \leftarrow$ Upsample(F_{13}, scale=2)

10: $F_{15} \leftarrow$ Concat(F_{14}, F_4)
11: $P_3 \leftarrow$ TAFHybridBlock(F_{15}, *C*_out=64)
12:
13: $F_{17} \leftarrow$ Conv(P_3, stride=2)
14: $F_{18} \leftarrow$ Concat(F_{17}, F_{13})
15: $P_4 \leftarrow$ TAFHybridBlock(F_{18}, *C*_out=128)
16:
17: $F_{20} \leftarrow$ Conv(P_4, stride=2)
18: $F_{21} \leftarrow$ Concat(F_{20}, F_0)
19: $P_5 \leftarrow$ TAFHybridBlock(F_{21}, *C*_out=256)
20:
21: # Detection Head
22: *D* ← Detect([P_3, P_4, P_5])
23: return *D*

Function: TAFHybridBlock(*X*, *C*_out)

Input: $X \in \mathbb{R}^{(C_in \times H \times W)}$
Output: $Y \in \mathbb{R}^{(C_out \times H \times W)}$

1: # Stem Convolution
2: *S* ← Conv(*X*, *C*_out, kernel=3, padding=1)
3: *S* ← SiLU(BatchNorm(*S*)) // [*C*_out, *H*, *W*]
4:
5: # ECA Branch: Efficient Channel Attention
6: *w*_eca ← AdaptiveAvgPool2d(*S*, 1) // [*C*_out, 1, 1]
7: *w*_eca ← Conv1d(*w*_eca, kernel=3)
8: *w*_eca ← Sigmoid(*w*_eca)
9: *F*_eca ← $S \odot w_eca$ // element-wise multiply
10:
11: # SE Branch: Squeeze-and-Excitation
12: *w*_se ← AdaptiveAvgPool2d(*S*, 1) // [*C*_out, 1, 1]
13: *w*_se ← SiLU(Conv(*w*_se, *C*_out//8, kernel=1))
14: *w*_se ← Sigmoid(Conv(*w*_se, *C*_out, kernel=1))
15: *F*_se ← $S \odot w_se$ // element-wise multiply
16:
17: # Fusion Layer
18: *F*_concat ← Concat([*F*_eca, *F*_se]) // [2×*C*_out, *H*, *W*]
19: *Y* ← Conv(*F*_concat, *C*_out, kernel=1)
20: *Y* ← SiLU(BatchNorm(*Y*))
21:
22: return *Y*

Table A2. Architectural specifications of evaluated variants: (a) DSC-YOLO, (b) MSDC-YOLO, (c) FRI-YOLO, (d) DAM-YOLO.

(a) DSC-YOLO			
Layer	**Type**	**Output Shape**	**Configuration**
0–10	Backbone	[256, 20 × 20]	Standard YOLOv11n
11	Upsample	[256, 40 × 40]	scale_factor = 2
12	Concat	[384, 40 × 40]	dim = 1, [11, 6]
13	DWTinyBlock	[128, 40 × 40]	DW k = 3 + PW k = 1
14	Upsample	[128, 80 × 80]	scale_factor = 2
15	Concat	[256, 80 × 80]	dim = 1, [14, 4]
16	DWTinyBlock	[64, 80 × 80]	DW k = 3 + PW k = 1
17	Conv	[64, 40 × 40]	k = 3, s = 2, p = 1
18	Concat	[192, 40 × 40]	dim = 1, [17, 13]
19	DWTinyBlock	[128, 40 × 40]	DW k = 3 + PW k = 1
20	Conv	[128, 20 × 20]	k = 3, s = 2, p = 1
21	Concat	[384, 20 × 20]	dim = 1, [20, 10]
22	DWTinyBlock	[256, 20 × 20]	DW k = 3 + PW k = 1
23	Detect	Multi-scale	[P3, P4, P5]
(b) MSDC-YOLO			
Layer	**Type**	**Output Shape**	**Configuration**
0–12	(Same as DSC)	[384, 40 × 40]	–
13	DilatedContext	[128, 40 × 40]	d = [1, 2, 3], k = 3
14–15	(Same as DSC)	[256, 80 × 80]	–
16	DilatedContext	[64, 80 × 80]	d = [1, 2, 3], k = 3
17–18	(Same as DSC)	[192, 40 × 40]	–
19	DilatedContext	[128, 40 × 40]	d = [1, 2, 3], k = 3
20–21	(Same as DSC)	[384, 20 × 20]	–
22	DilatedContext	[256, 20 × 20]	d = [1, 2, 3], k = 3
23	Detect	Multi-scale	[P3, P4, P5]
(c) FRI-YOLO			
Layer	**Type**	**Output Shape**	**Configuration**
0–22	(Standard Neck)	–	Same as YOLOv11n
16 *	LinearLite	[64, 80 × 80]	1 × 1 Conv
19 *	LinearLite	[128, 40 × 40]	1 × 1 Conv
22 *	LinearLite	[256, 20 × 20]	1 × 1 Conv
23	Detect	Multi-scale	[P3 *, P4 *, P5 *]
Note: * indicates refined features via LinearLite interface.*			
(d) DAM-YOLO			
Layer	**Type**	**Output Shape**	**Configuration**
0–12	(Same as DSC)	[384, 40 × 40]	–
13	TAFHybrid	[128, 40 × 40]	ECA (k = 3) + SE (r = 8)
14–15	(Same as DSC)	[256, 80 × 80]	–
16	TAFHybrid	[64, 80 × 80]	ECA (k = 3) + SE (r = 8)
17–18	(Same as DSC)	[192, 40 × 40]	–
19	TAFHybrid	[128, 40 × 40]	ECA (k = 3) + SE (r = 8)
20–21	(Same as DSC)	[384, 20 × 20]	–
22	TAFHybrid	[256, 20 × 20]	ECA (k = 3) + SE (r = 8)
23	Detect	Multi-scale	[P3, P4, P5]

Legend: k: kernel size. s: stride. p: padding. d: dilation rate. r: reduction ratio (SE attention). DW: depthwise convolution. PW: pointwise convolution.

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